Q1:2006 SPEC P2

6 (a) Explain what is meant by *frequency modulation* (FM).

....

....

.... [2]

(b) A sinusoidal carrier wave has amplitude 12 V and frequency 600 kHz. The frequency of the carrier wave changes by 25 kHz per volt.

The carrier wave is used for the transmission of a signal of frequency 3.0 kHz and amplitude 2.0 V.

For the frequency modulated carrier wave, state

(i) the amplitude,

amplitude = ... V [1]

(ii) the maximum frequency,

maximum frequency = ... kHz [1]

(iii) the minimum frequency,

minimum frequency = ... kHz [1]

(iv) the number of times per second that the frequency of the carrier wave changes from the maximum value, to the minimum value and then back to the maximum value.

number = ... [1]

7 Before the development of microwave links and optic fibres, co-axial cables were used widely for telephone communication.

(a) Fig. 16.1 shows one type of co-axial cable.

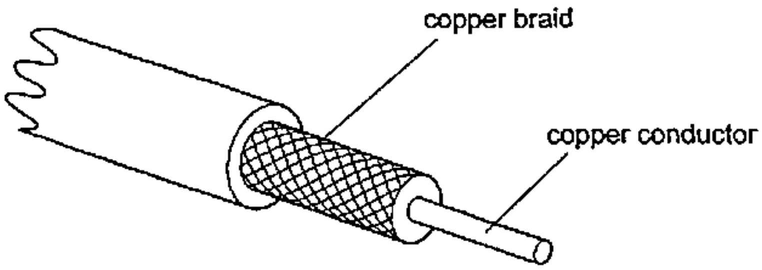

Fig. 16.1

State the purpose of the copper braid and how this purpose is achieved.

....

....

.... [2]

(b) One advantage over co-axial cables of microwave links and of optic fibres is increased bandwidth.

Explain why increased bandwidth has led to a reduction in the cost of telephone calls.

....

....

....

.... [3]

Q2:JUN 2019 P2

6 (a) Explain why there is signal attenuation in

(i) communication wires.

____

____

____

(ii) air.

____

____

(iii) optical fibres.

____

____

____ [3]

6032/2 J2019

(b) A signal of power 15 mW is transmitted in a cable whose attenuation is 15 dB. The cable is connected to another cable which has an attenuation of 25 dB.

Calculate the power transmitted at the end of the second cable.

[4]

(c) **Fig. 6.1** shows the variation of displacement of an amplitude modulated radio wave with time.

Fig. 6.1

Calculate the frequency of the

(i) carrier wave,

frequency ___________

(ii) information signal.

frequency ___________ [3]

Q3:NOV 2018 P2

10 **(a)** **(i)** Define the terms

1. *amplitude modulation*,

____

____

____

2. *bandwidth*.

____

(ii) Explain why

1. a larger bandwidth allows a higher transmission,

____

____

____

2. a higher transmission rate allows a more faithful reproduction of information.

____

____

____ [6]

(b) A carrier of frequency 800 kHz is amplitude modulated by frequencies ranging from 1 kHz to 10 kHz.

Determine the frequency range covered by each sideband.

____ [2]

Q4:JUN 2019 P3

5 (c) (i) Give **three** advantages for transmitting data in digital form.

(ii) **Fig. 5.3** shows an analogue output of a microphone.

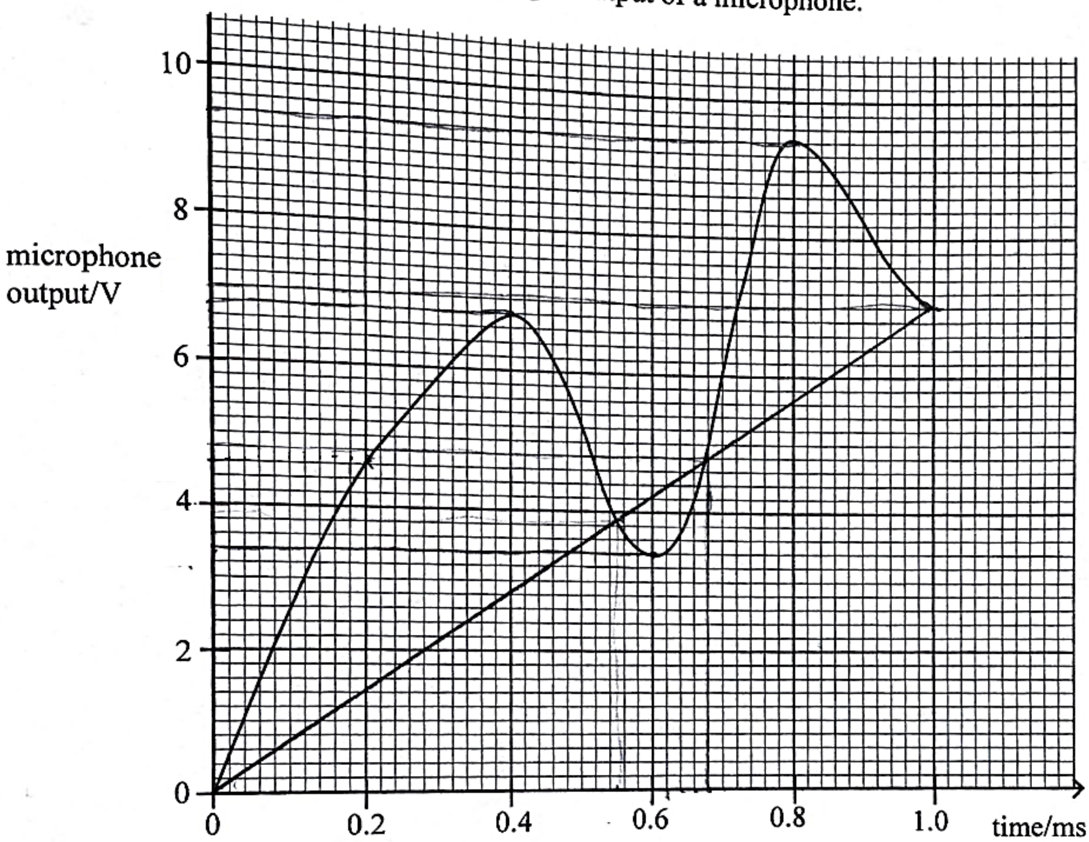

Fig. 5.3

The output is sampled at 0.2 ms intervals.

1. Determine the sampling rate.

2. Use the four-bit system to represent the output of the microphone in digital form at each interval.

3. Suggest **two** ways of improving the matching between the microphone output and its digital form.

[12]

Q5:NOV 2018 P3

5 (a) (i) Define a *carrier wave* and state the advantage of its use.

(ii) Distinguish between amplitude modulation and frequency modulation.

(iii) Fig.5.1 shows a modulated wave.

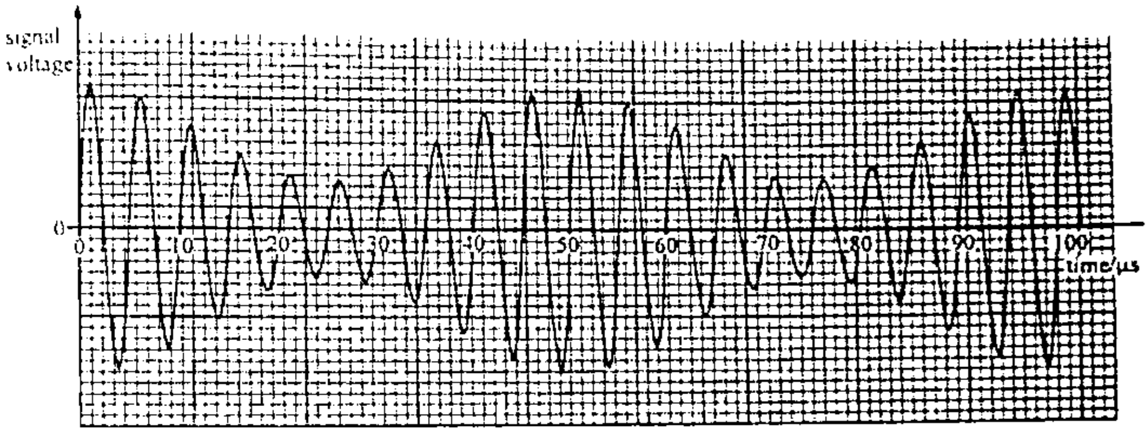

Fig.5.1

Using **Fig.5.1**

1. draw the frequency spectrum of the modulated wave,

2. determine the bandwidth. **[8]**

(b) (i) State what is meant by a *digital signal.*

(ii) Explain how transmission of data in digital form is an advantage over the transmission of data in analogue form.

(iii) Describe how the accuracy of the reproduction of the original analogue signal can be improved in analogue to digital conversion. [5]

(c) (i) State any **two** advantages of using geostationary satellites.

(ii) In satellite communication

1. state how swamping is overcome.

2. why the satellites are used in conjunction with optical cables. **[3]**

(d) (i) Define *attenuation*.

(ii) State the advantage of using the decibel in comparisons.

(iii) The input power to a cable of length 40 km is 400 mW. The background noise is 3.0×10^{-13} W and the minimum signal to noise ratio permissible is 15 dB. The attenuation per unit length in the cable is 1.6 dB km^{-1}.

Calculate the

1. power output from the cable,

2. maximum interrupted length of cable along which the signal can be transmitted. **[6]**

(e) State the type of radio waves with frequency

1. between 3 MHz – 3 MHz,

2. greater than 30 MHz. **[2]**

Q6:2018 SPEC P3

5 **(a)** **Fig. 5.1** shows a coaxial cable.

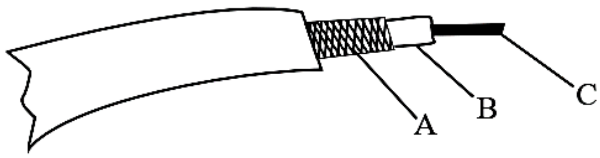

Fig. 5.1

(i) State the

1. names of the components labeled A, B and C,

2. functions of component A.

(ii) List **three** advantages of coaxial cables over wire pairs.

[8]

(b) **(i)** Define the term *modulation*.

(ii) Explain why

1. the reception of amplitude modulated signals is poorer than the frequency modulated signals,

2. frequency modulated transmissions are more expensive than amplitude modulated transmissions.

[6]

(c) **Fig. 5.2** shows an original analogue signal before transmission.

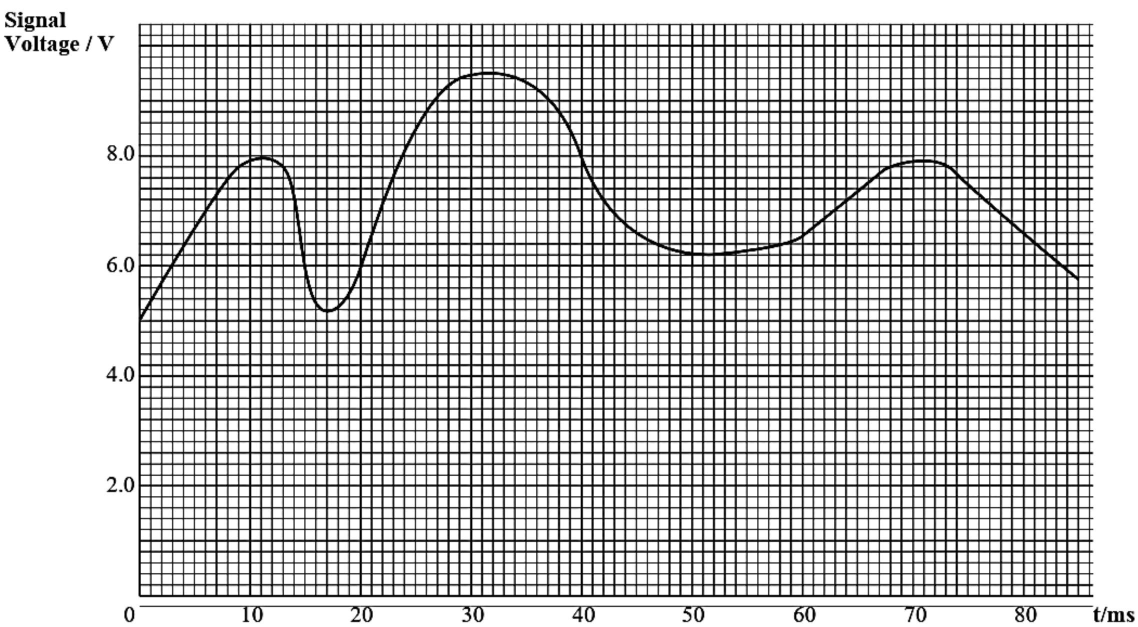

Fig. 5.2

(i) Copy **Table 5.1** and complete it using **Fig. 5.2**.

Table 5.1

Time/μs	0	40	70
Voltage/V	5.0		
Digital code	0101		

(ii) Using a sampling frequency of 100 kHz,

1. reconstruct **Fig. 5.2** to produce the **"recovered"** analogue signal after transmission and sketch it.

2. Explain how the reconstruction can be improved to have a more accurate representation of the original signal.

[11]

Q7:2018 SPEC P2

9. **(a)** Explain why it is much more difficult to reduce the effects of noise when transmitting an analogue signal than when transmitting in digital form.

____

____

____

____ [2]

(b) **(i)** State **two** advantages of optical fibre cables over copper cables.

____

____

____

(ii) Explain the term *attenuation*.

____

____

(iii) State **one** cause of *attenuation* in optical fibres.

____

____[4]

(c) **(i)** Distinguish between a *geostationary* satellite and a *polar* satellite.

____

____

____

____

(ii) Give **three** reasons why microwaves are used for satellite communication.

____

____

____ [5]